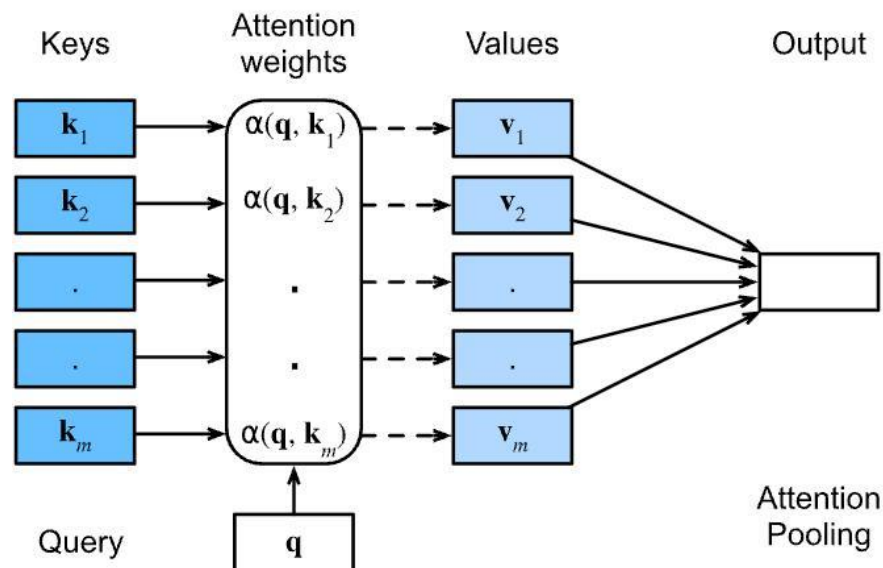


Attention Mechanism & Self-Attention:



RNNs struggle with **long-range dependencies**

Attention allows model to **focus on relevant words**

Attention Mechanism (General)

Given:

- Query (Q)
- Key (K)
- Value (V)

Formula

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Computes importance of each word

Self-Attention

Same sequence used for Q, K, V

Example:

“The cat sat on the mat”

“sat” attends to “cat”

Steps

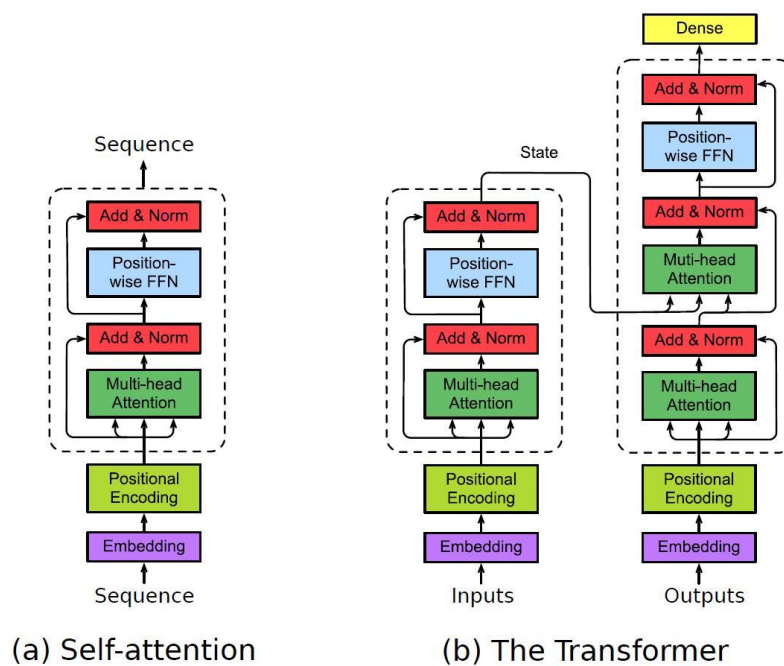
1. Compute Q, K, V

2. Compute similarity scores
3. Apply softmax
4. Weighted sum → output

Advantages

- Captures global dependencies
- Parallel computation
- No recurrence

Transformer & BERT Architecture:



Transformer (Vaswani et al., 2017)

Architecture

- Encoder (stack of layers)
- Decoder (stack of layers)

Each layer:

1. Multi-head attention
2. Feedforward network
3. Residual + normalization

Encoder Block

$$\text{Output} = \text{FFN}(\text{Attention}(X))$$

BERT (Bidirectional Encoder Representations from Transformers)

Uses **only encoder**

Key Features

- Bidirectional context
- Pre-trained model

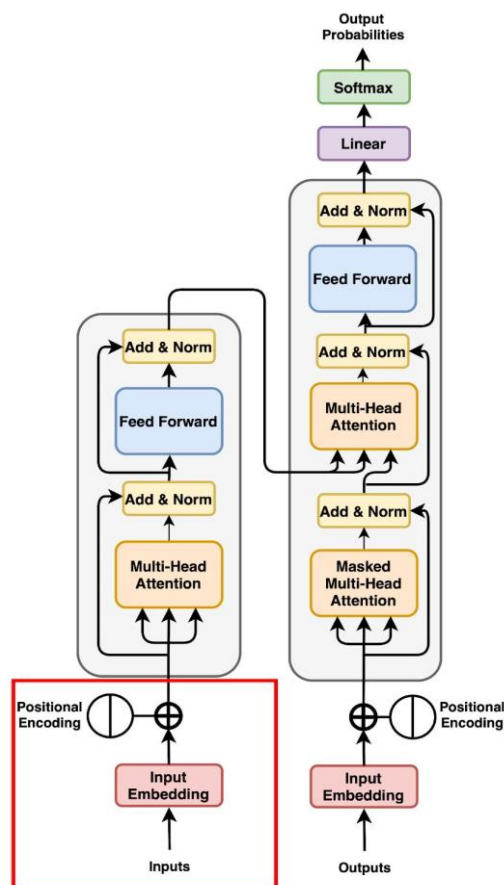
Input Format

[CLS] sentence [SEP]

Training Tasks

- Masked Language Model (MLM)
- Next Sentence Prediction (NSP)

Positional Encoding:



Problem

Transformers have **no notion of order**

Solution

Add positional information

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$
$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

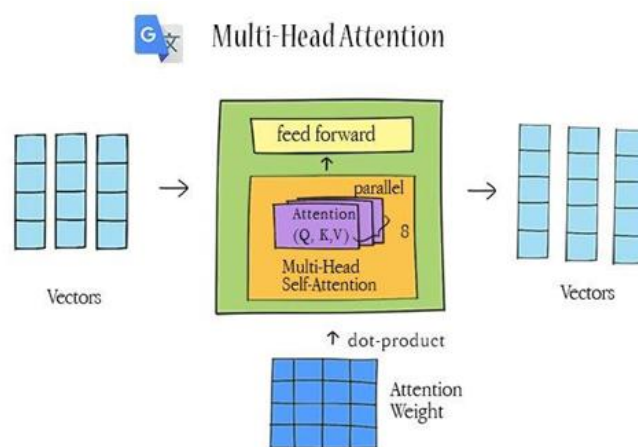
Key Idea

Each position has unique encoding

Sine/Cosine is used:

- Smooth representation
- Generalizes to longer sequences

Multi-Head Attention:



Run multiple attention operations in parallel

Formula

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

Each head:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

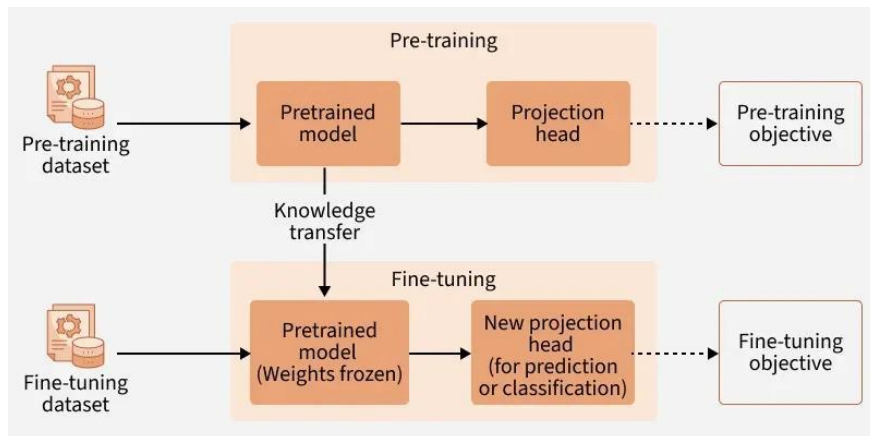
Multiple Heads are used because

- Capture different relationships
- Improves representation

Example

- One head → syntax
- One head → semantics

Pre-trained Language Models & Fine-Tuning:



Train model on large data → reuse for tasks

Pre-training

- Learn general language patterns

Fine-tuning

- Adapt model to specific task

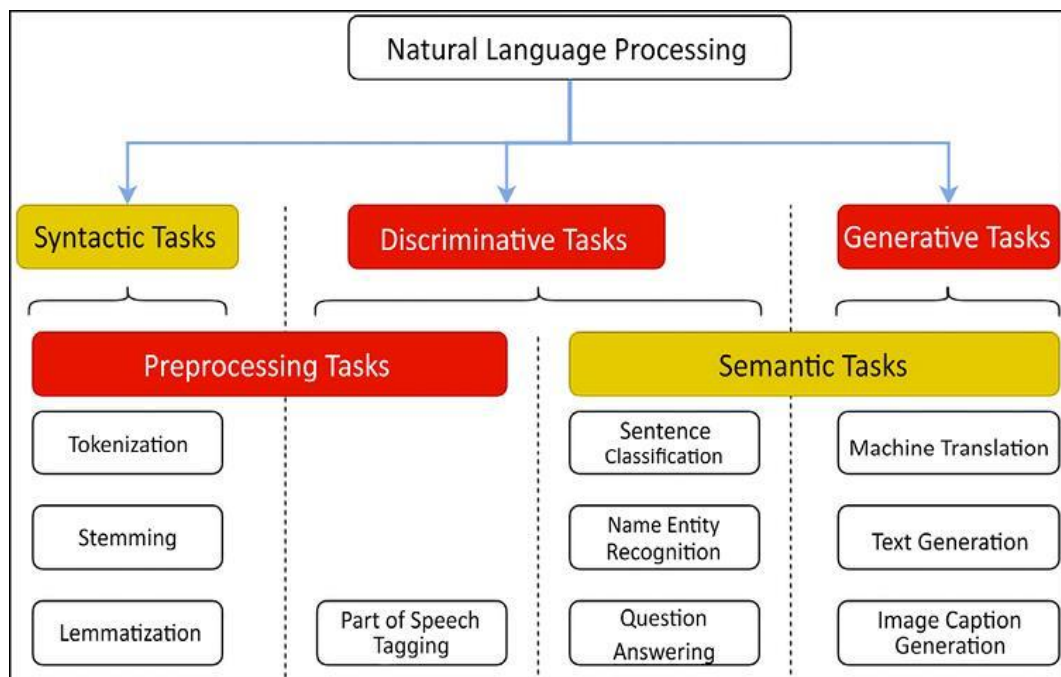
Advantages

- Less data required
- Faster training
- Better performance

Examples

- BERT
- GPT
- RoBERTa

Applications in NLP (Text Classification & Translation)



Text Classification

- Sentiment analysis
- Spam detection

Process

Input → Transformer → Classification head

Machine Translation

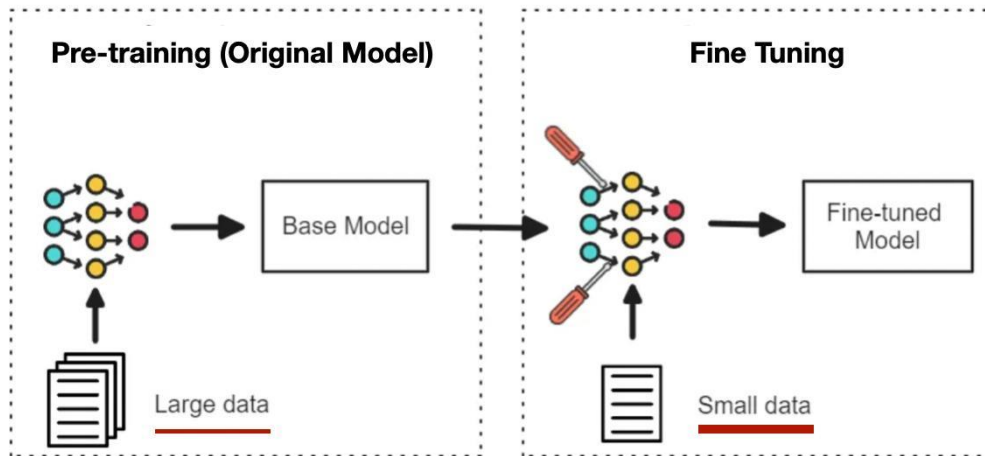
- English → French
- Uses encoder-decoder

Other Applications

- Question answering
- Named Entity Recognition

Large Language Models & Transfer Learning:

Large Language Model



Large Language Models (LLMs)

Very large transformer models trained on huge datasets

Examples

- GPT-4
- LLaMA

Key Features

- Billions of parameters
- Context understanding
- Text generation

Transfer Learning

Concept

Reuse knowledge from one task to another

Types

- Feature extraction
- Fine-tuning

Benefits

- Reduces training cost
- Works with small data

